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(54) **SEMICONDUCTOR DEVICE STRUCTURE
AND METHODS OF FORMING THE SAME**

(71) Applicant: **TAIWAN SEMICONDUCTOR
MANUFACTURING COMPANY,
LTD.**, Hsinchu (TW)

(72) Inventors: **Chao-Hsuan CHEN**, Hsinchu (TW);
I-Wei YANG, Yilan (TW); **Wen-Shuo
HSIEH**, Taipei (TW); **Shu-Yuan KU**,
Hsinchu (TW); **Ryan Chia-Jen CHEN**,
Hsinchu (TW)

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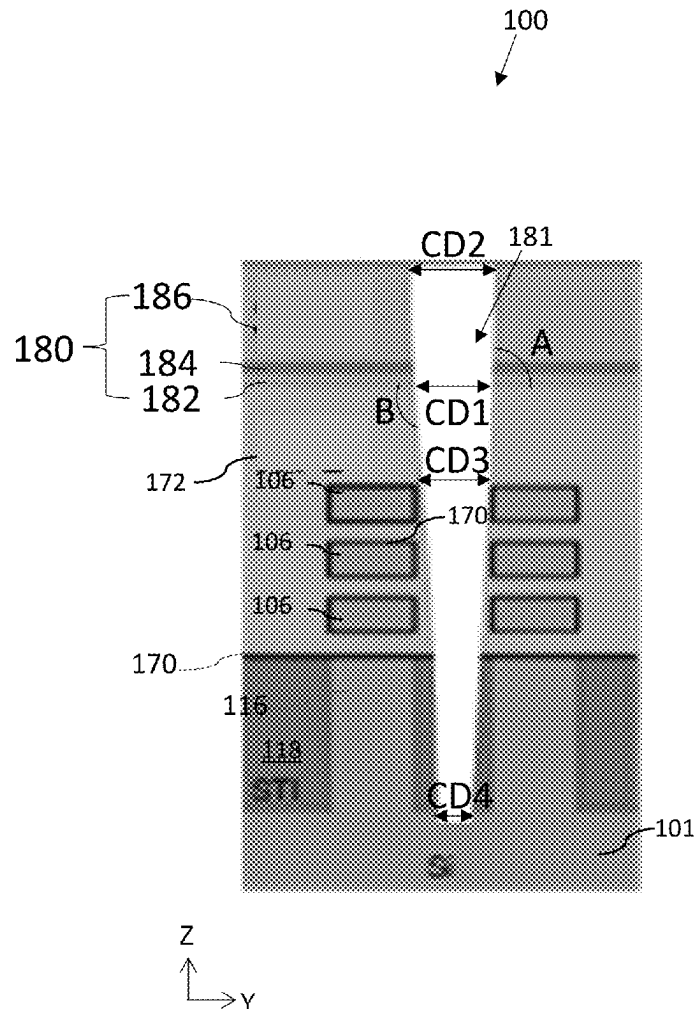
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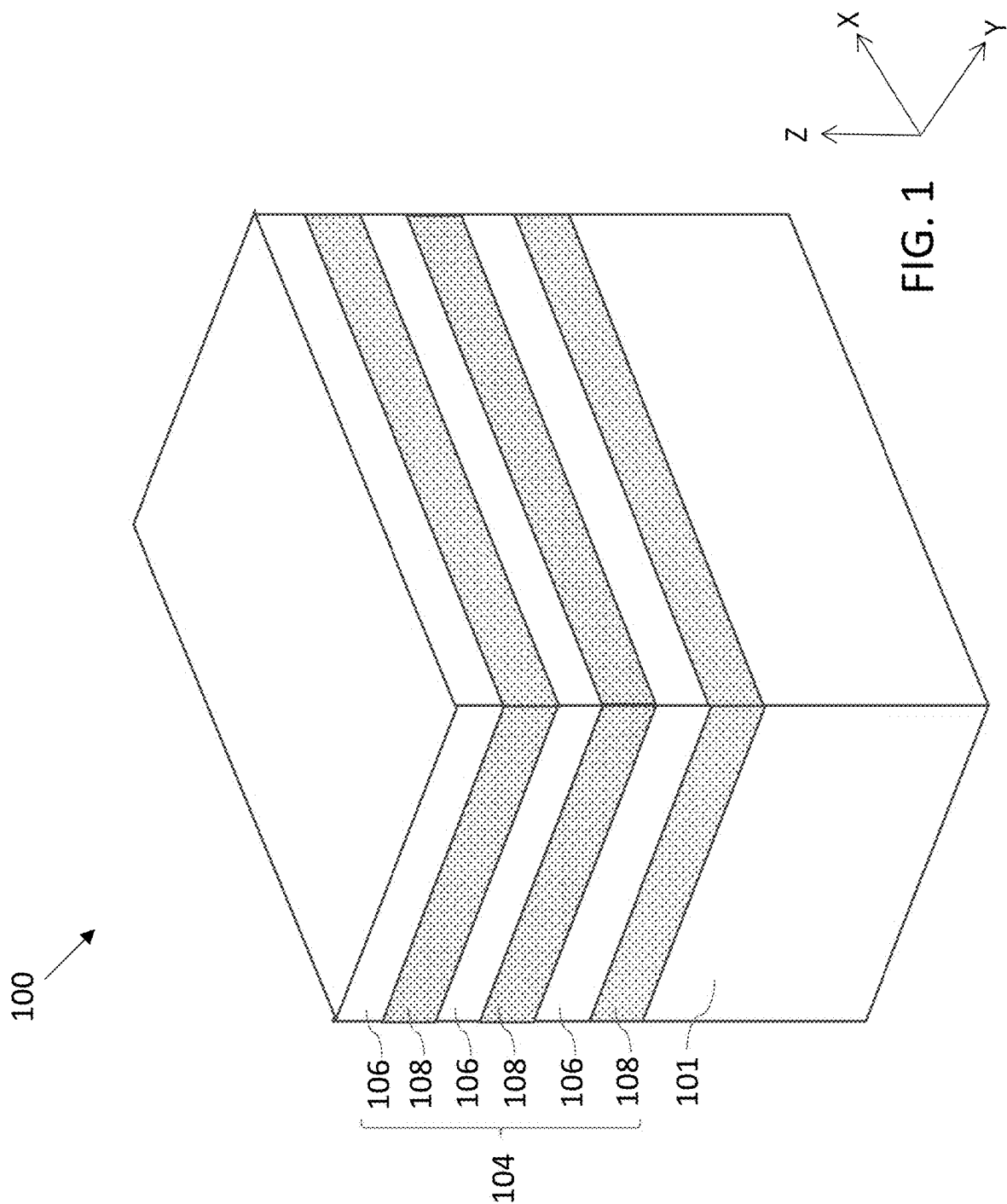
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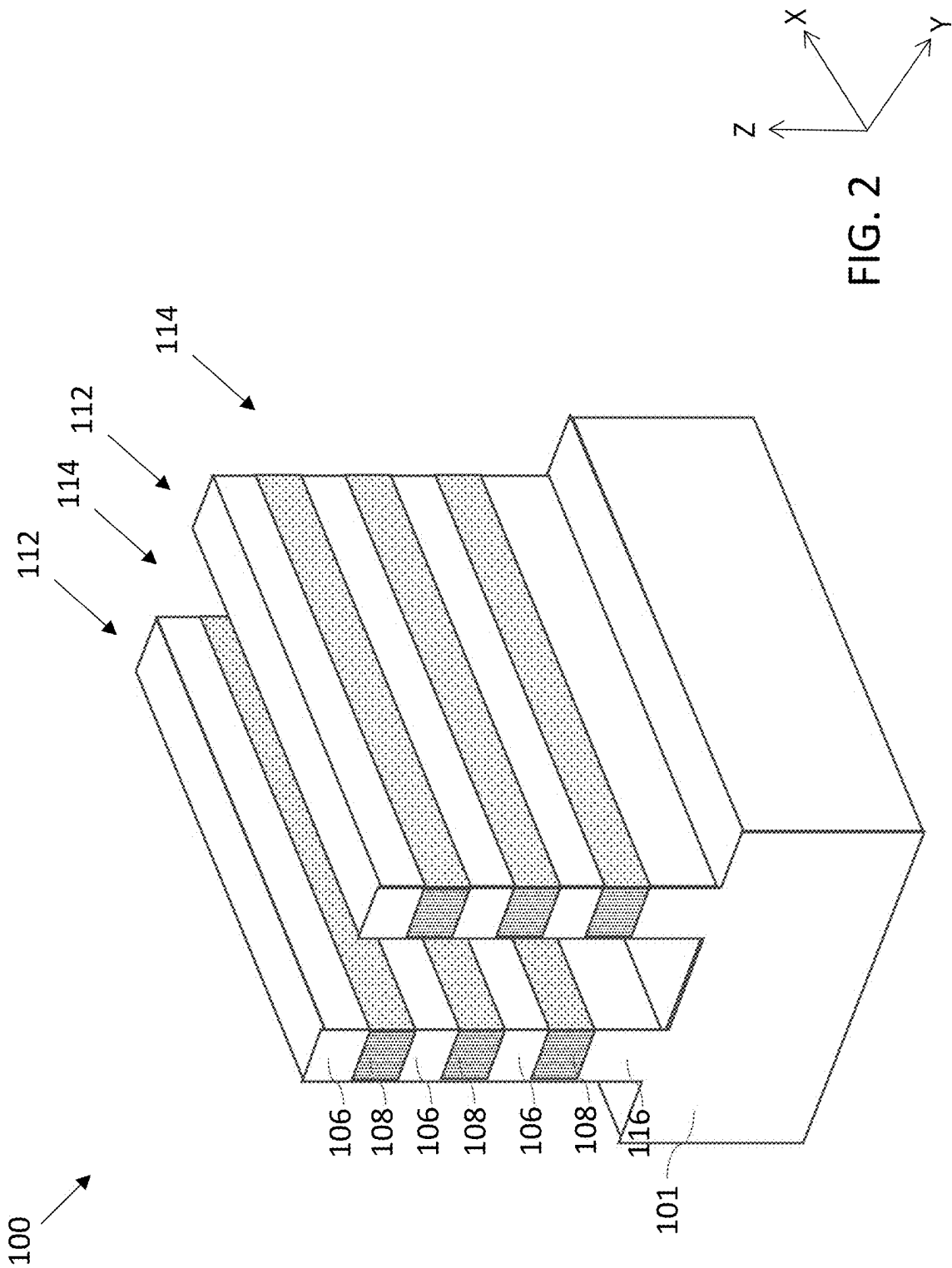
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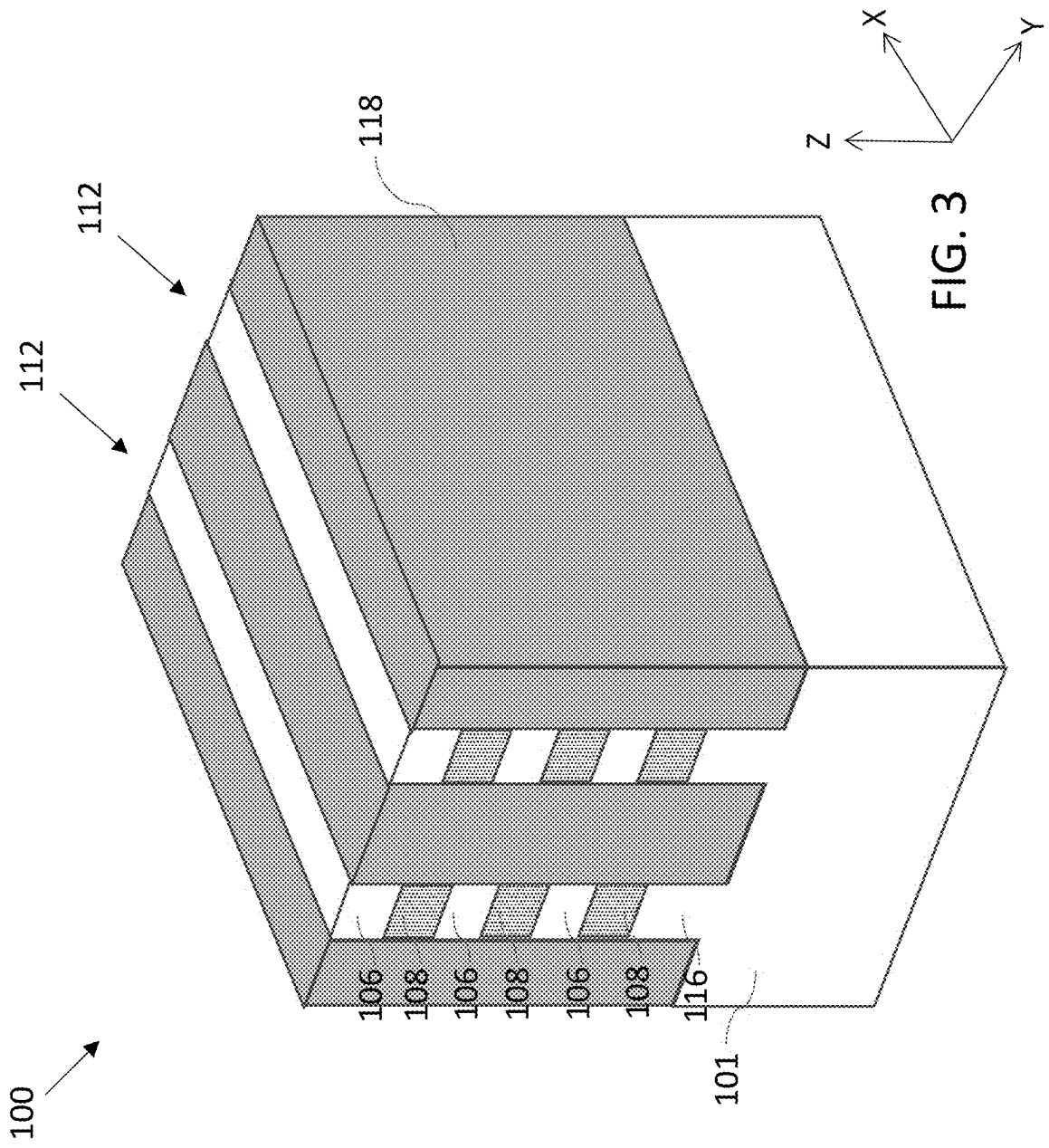
(57) **ABSTRACT**

Embodiments of the present disclosure provide a method for forming a semiconductor device structure. The method includes forming a fin structure over a substrate, forming an insulating material adjacent the fin structure, depositing a gate dielectric layer over the fin structure, depositing a gate electrode layer on the gate dielectric layer, and forming an opening through the gate electrode layer and the gate dielectric layer into the insulating material by a cyclic process. The cyclic process includes performing a first number of cycles, each cycle of the first number of cycles includes a first main etch process having a first plasma bias voltage, performing a second number of cycles, and each cycle of the second number of cycles includes a second main etch process having a second plasma bias voltage substantially greater than the first plasma bias voltage. The method further includes filling the opening with a dielectric material.









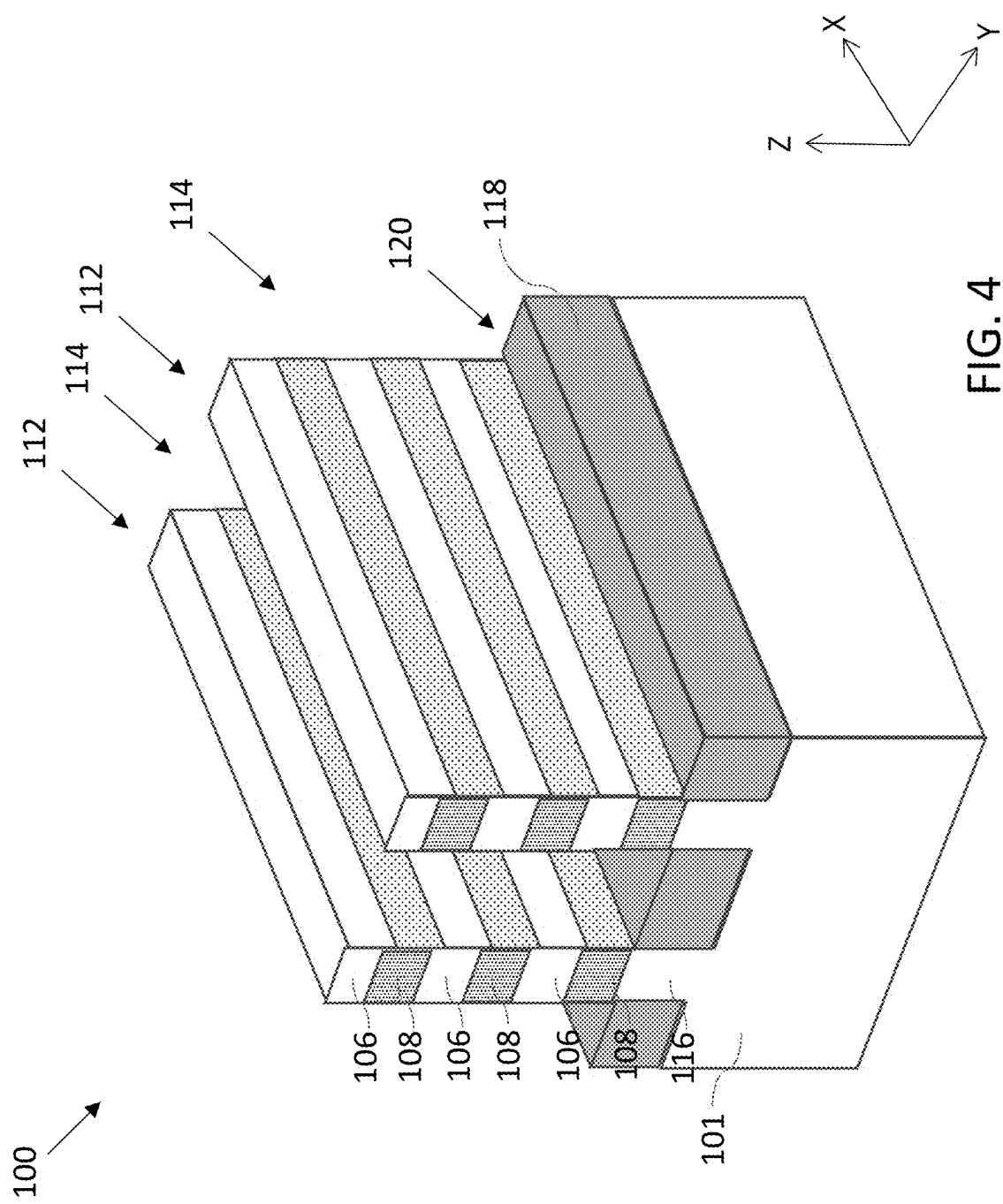
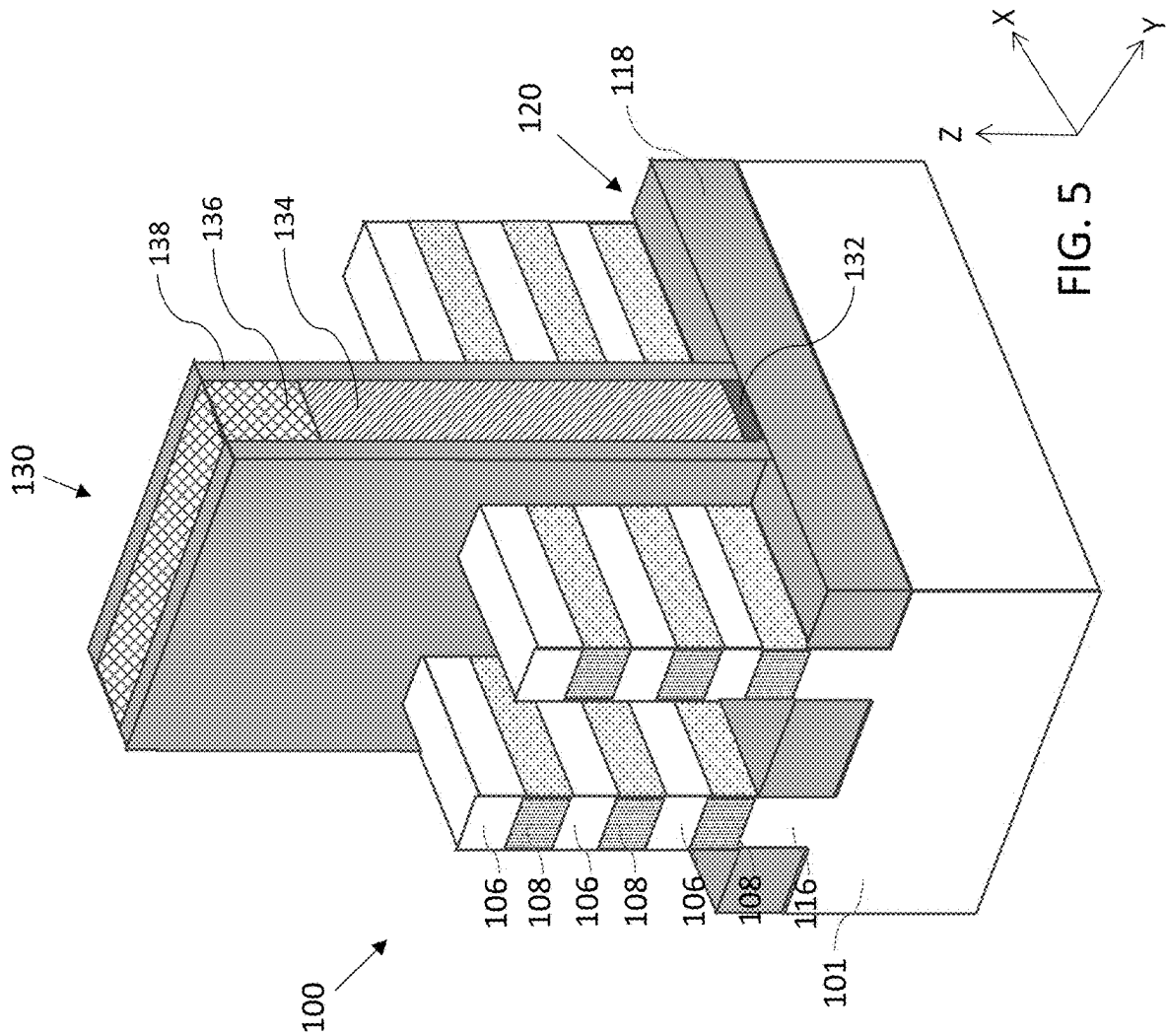
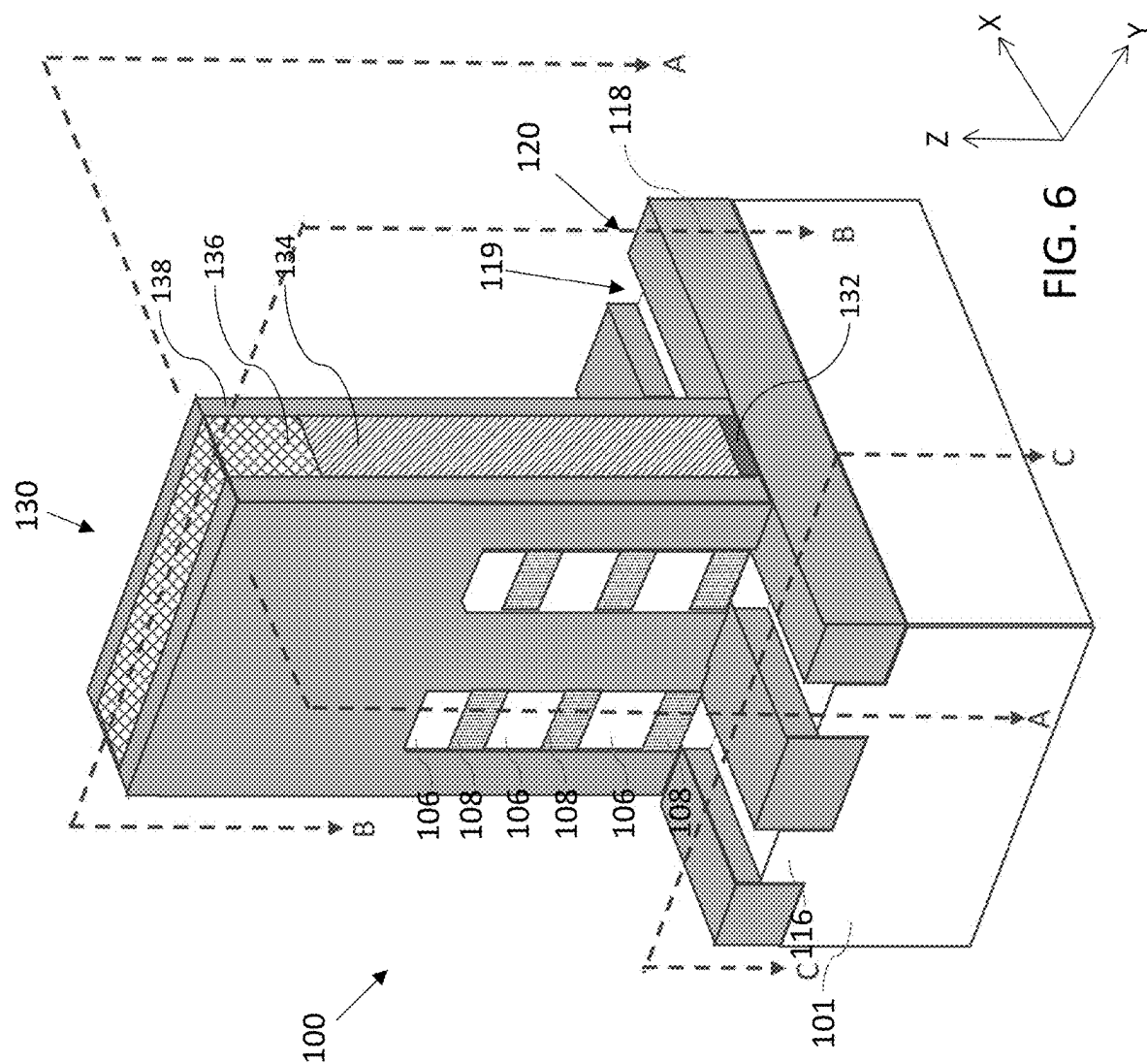
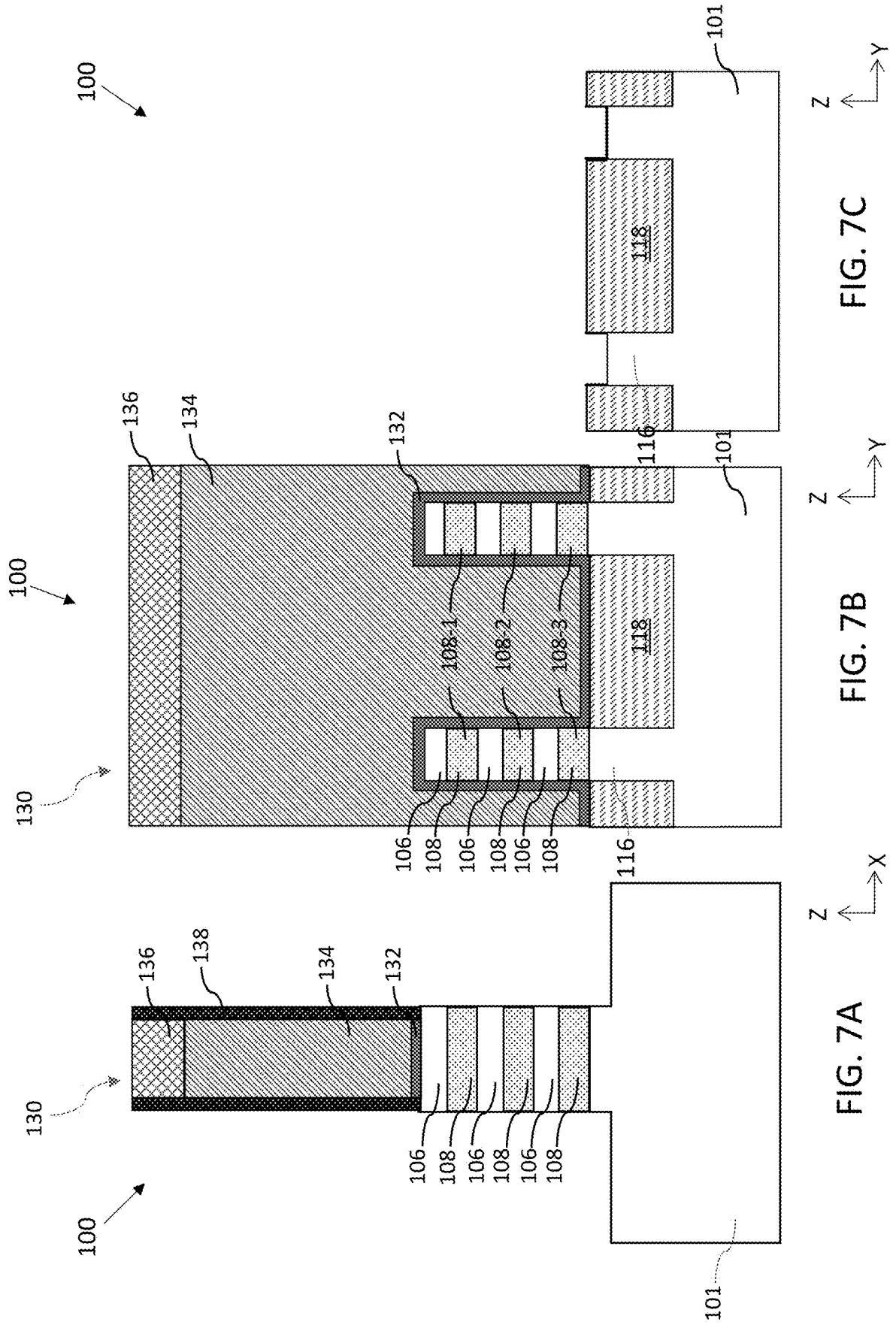
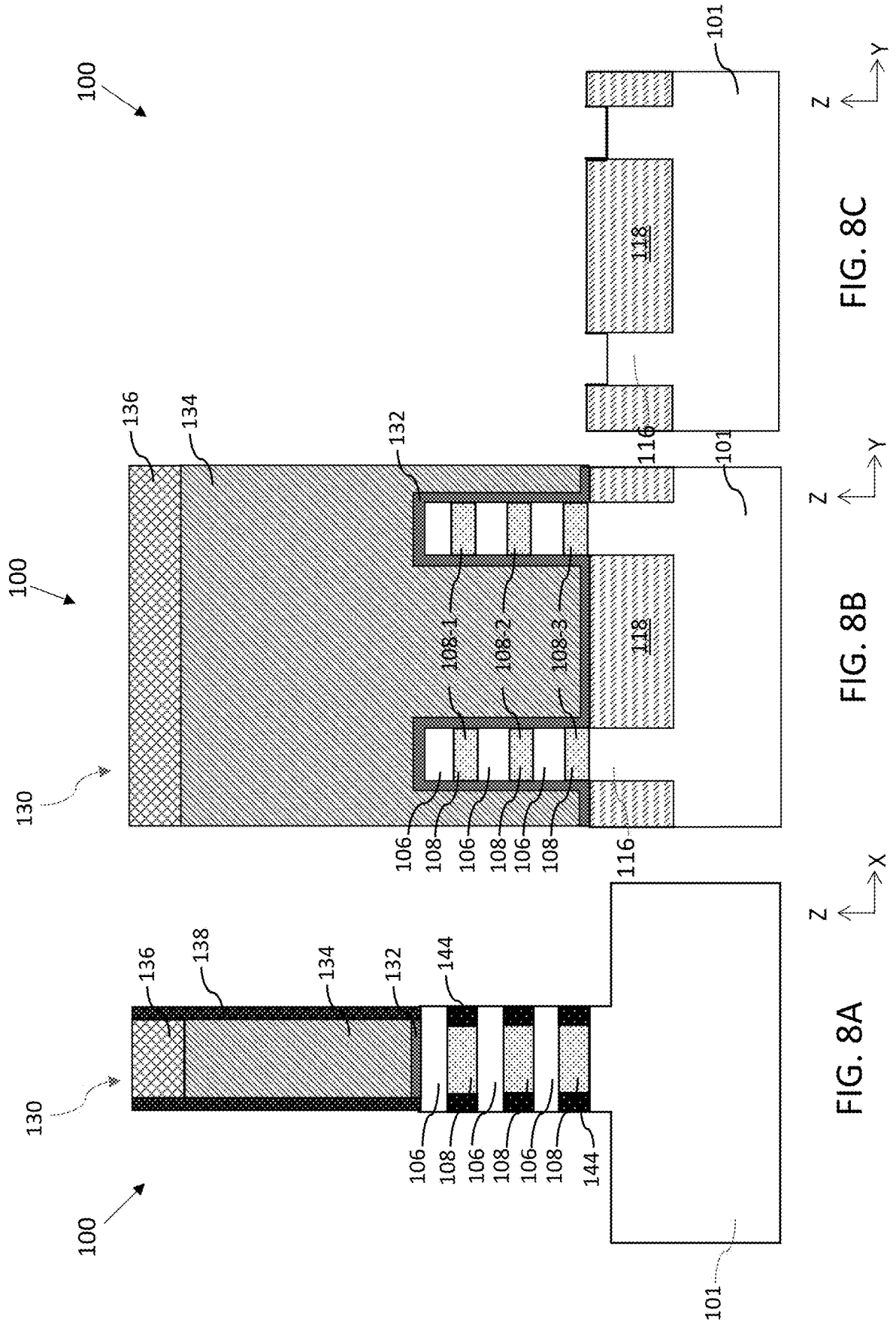


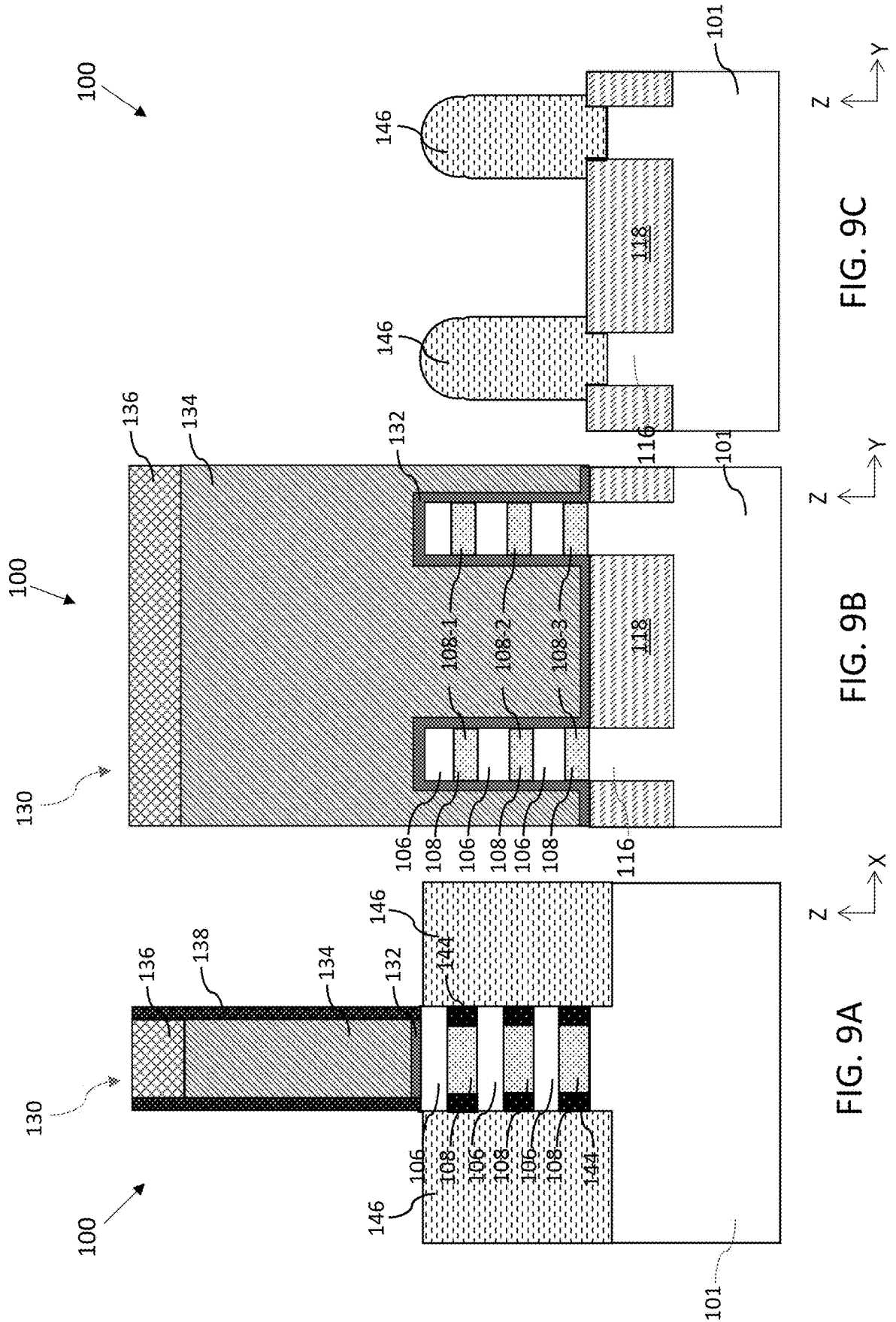
FIG. 4

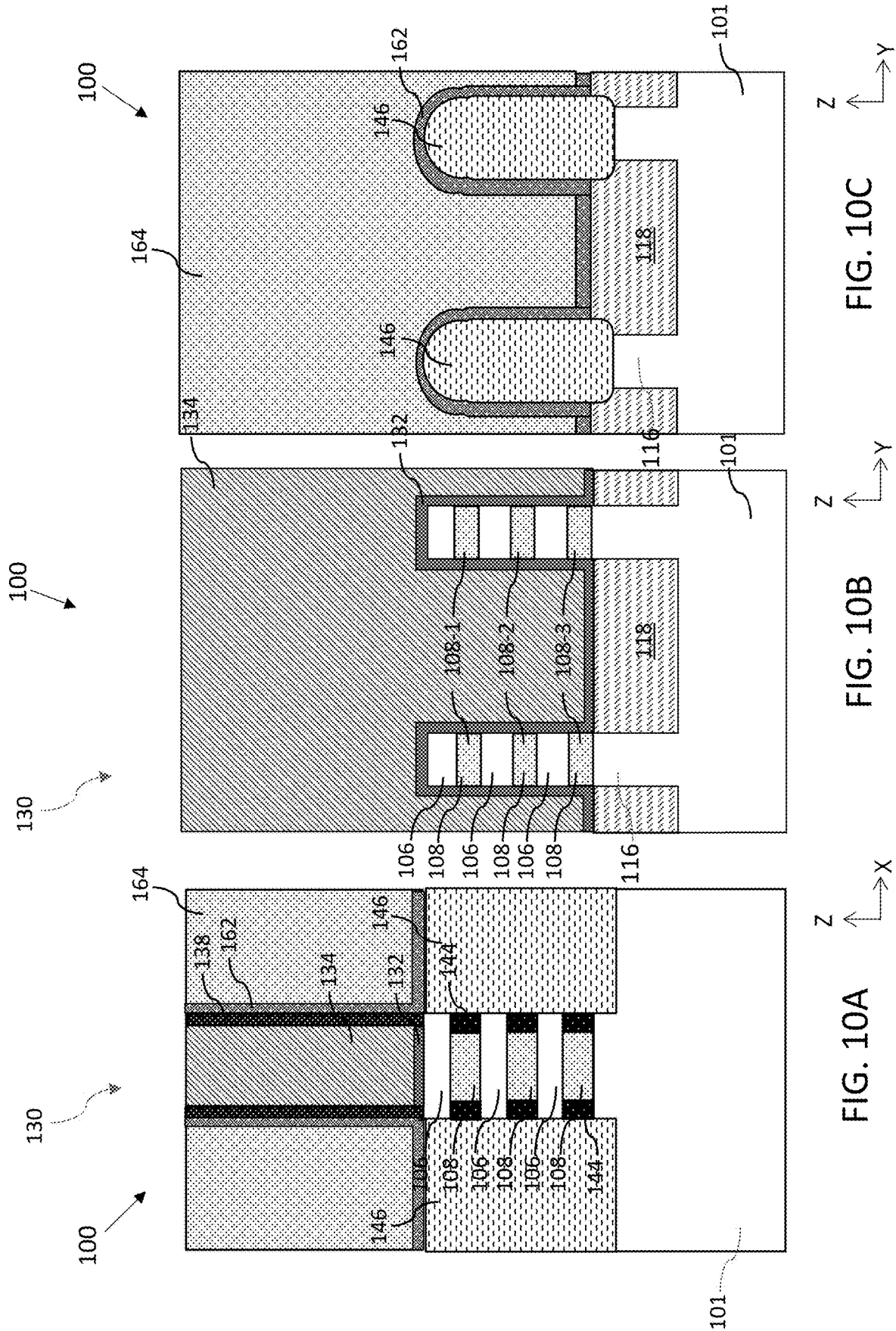


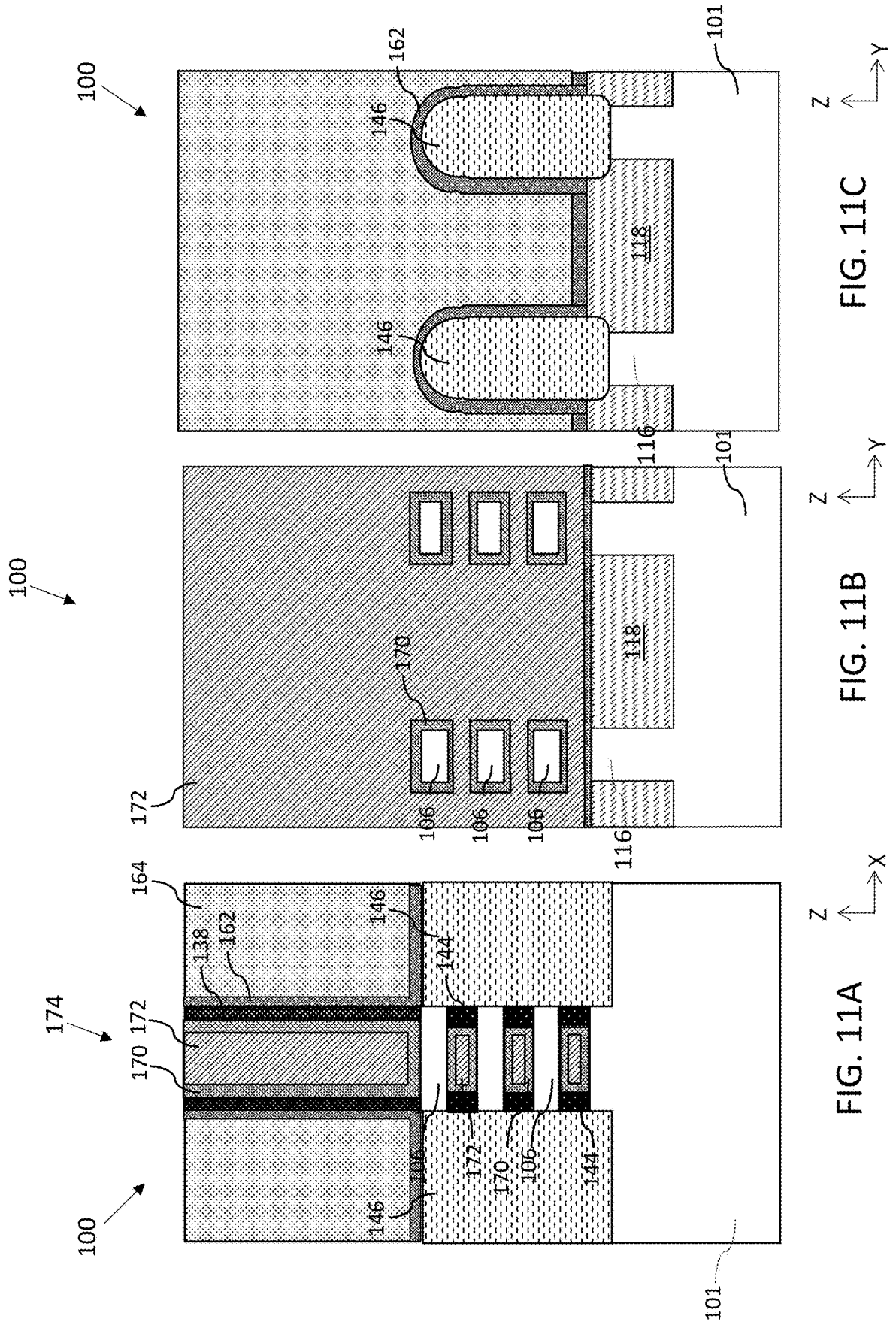


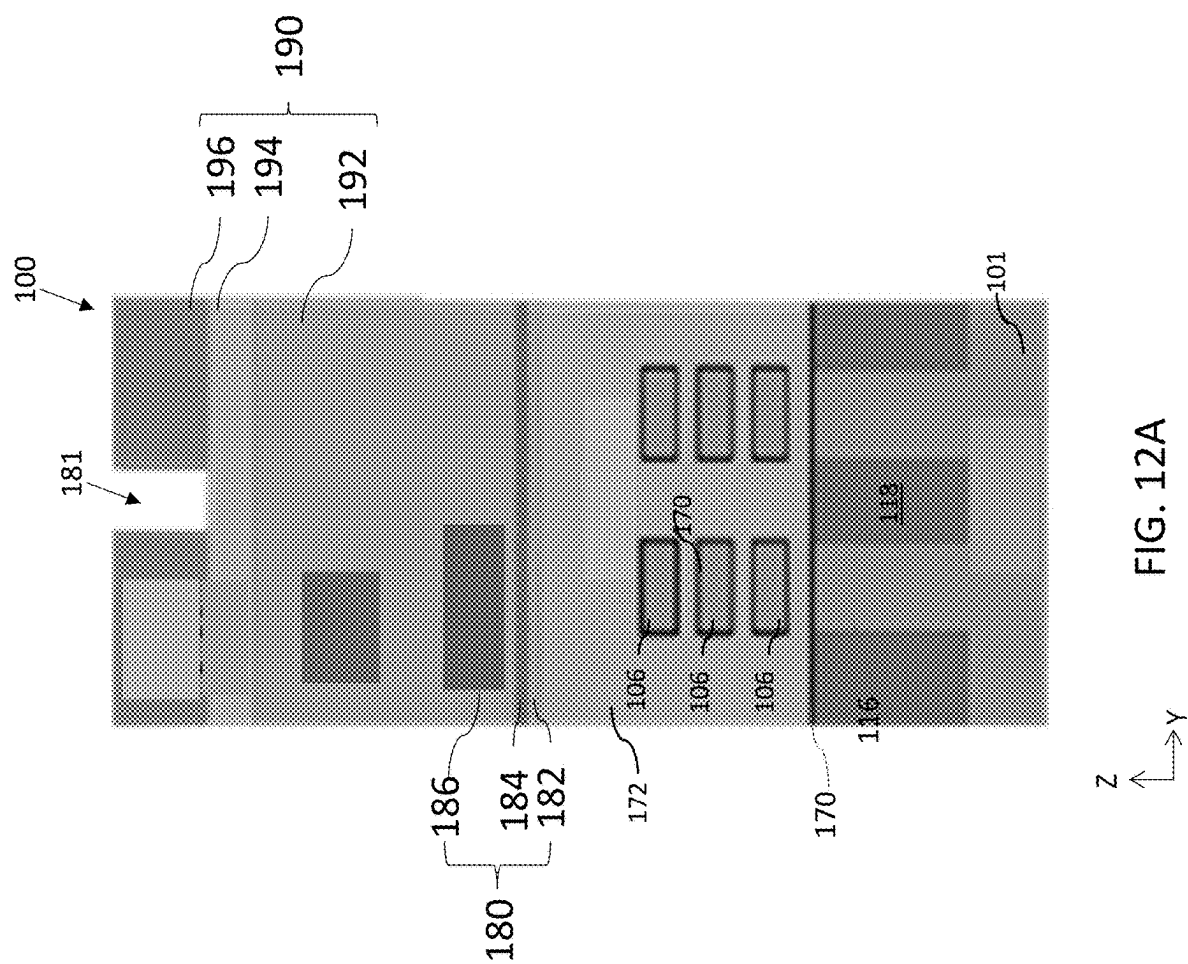


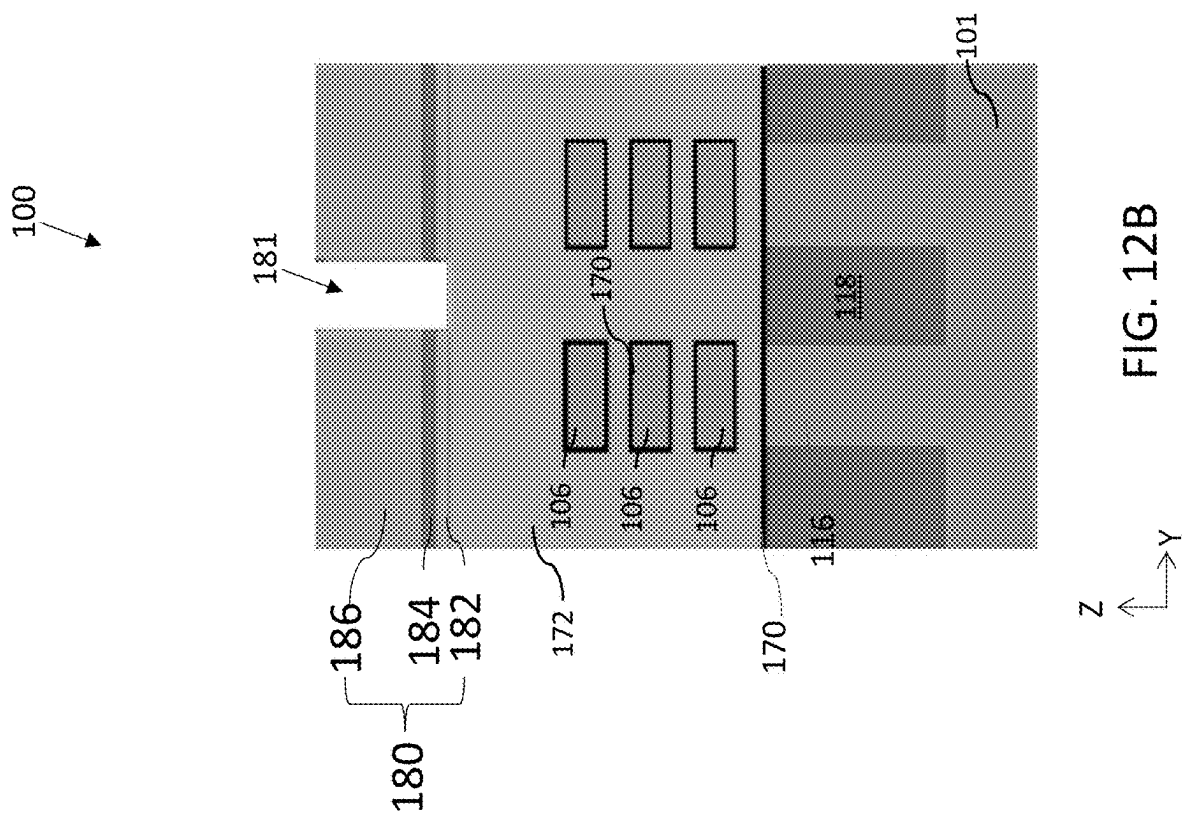


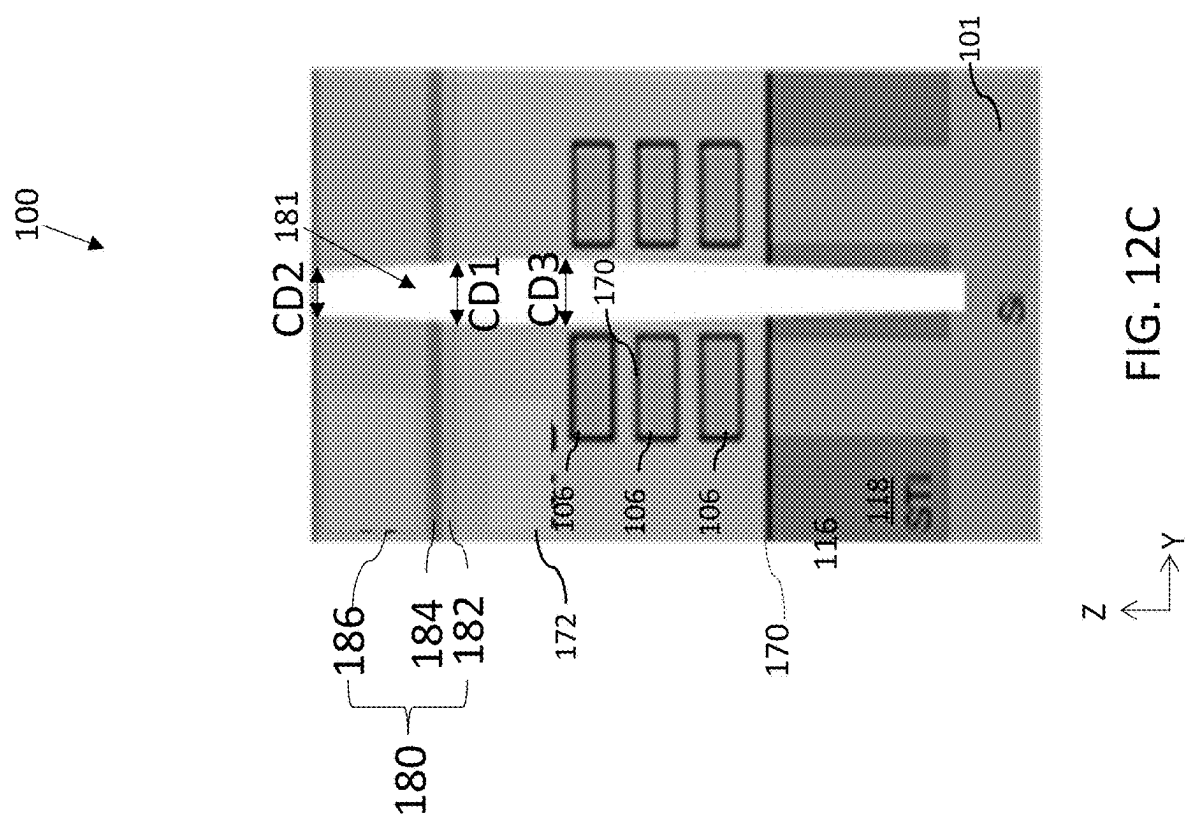


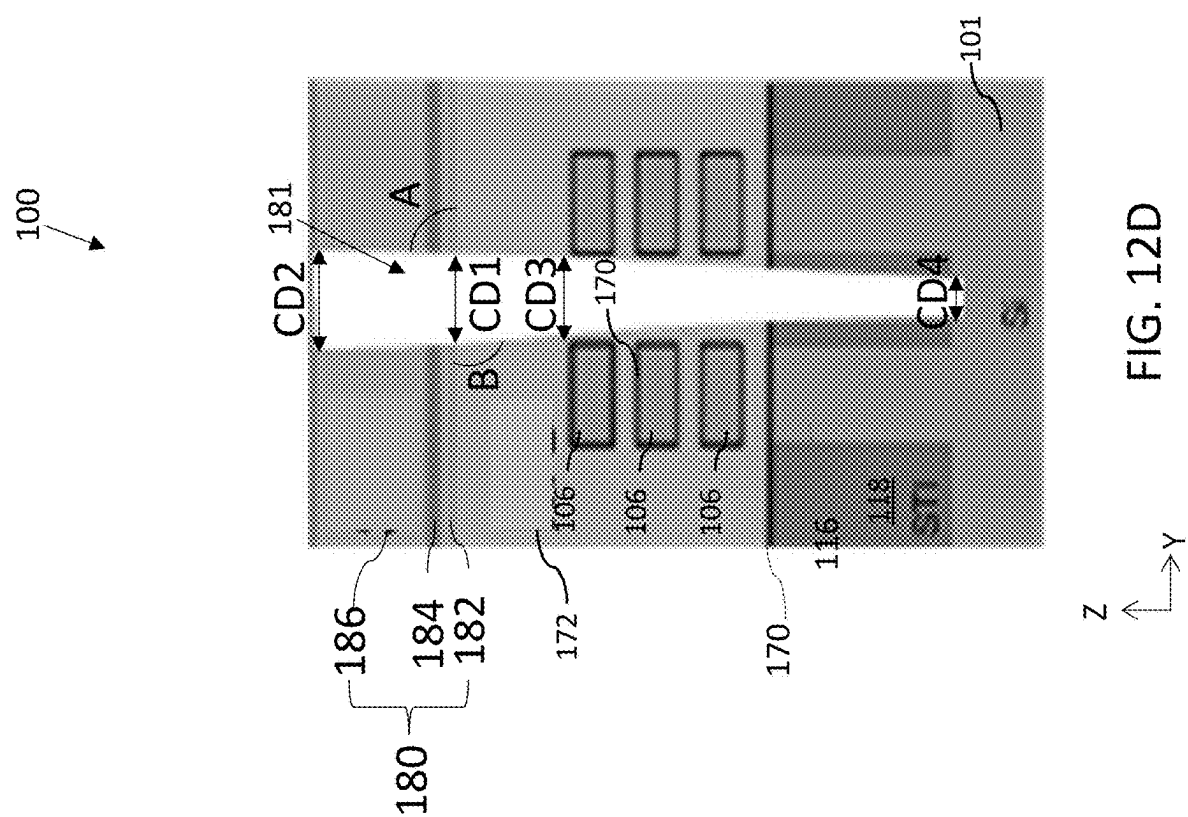












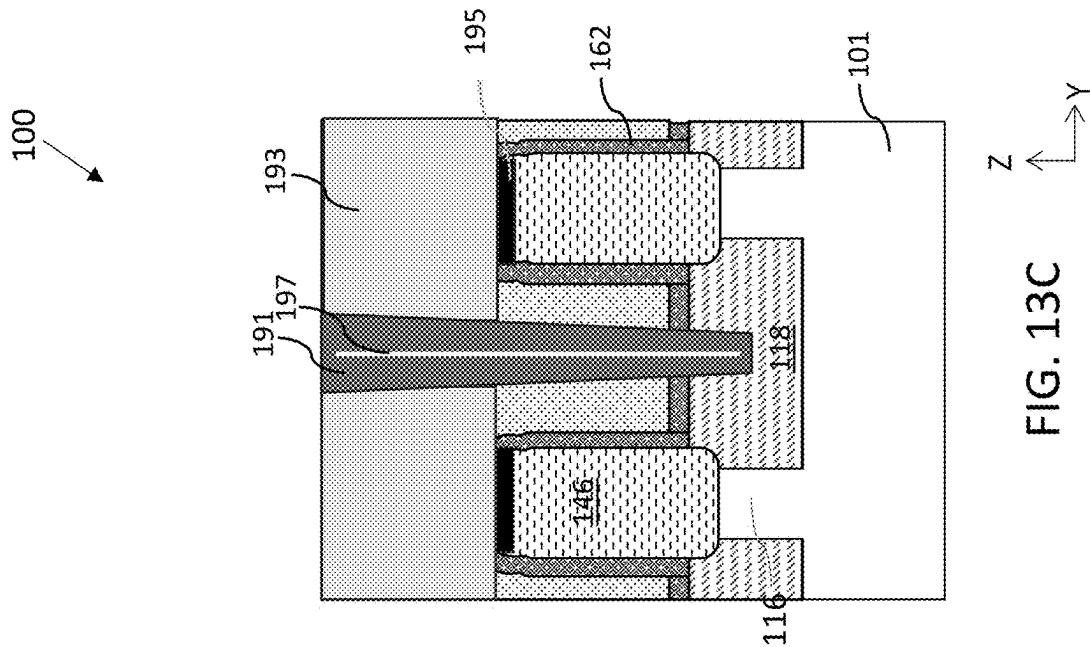


FIG. 13C

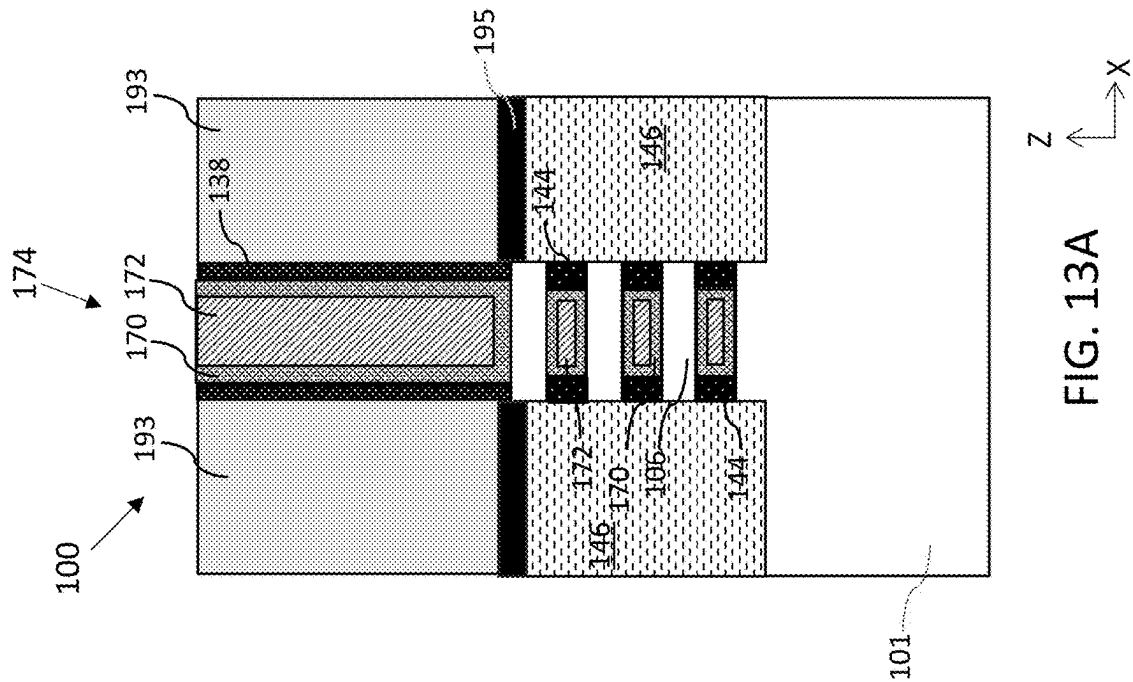
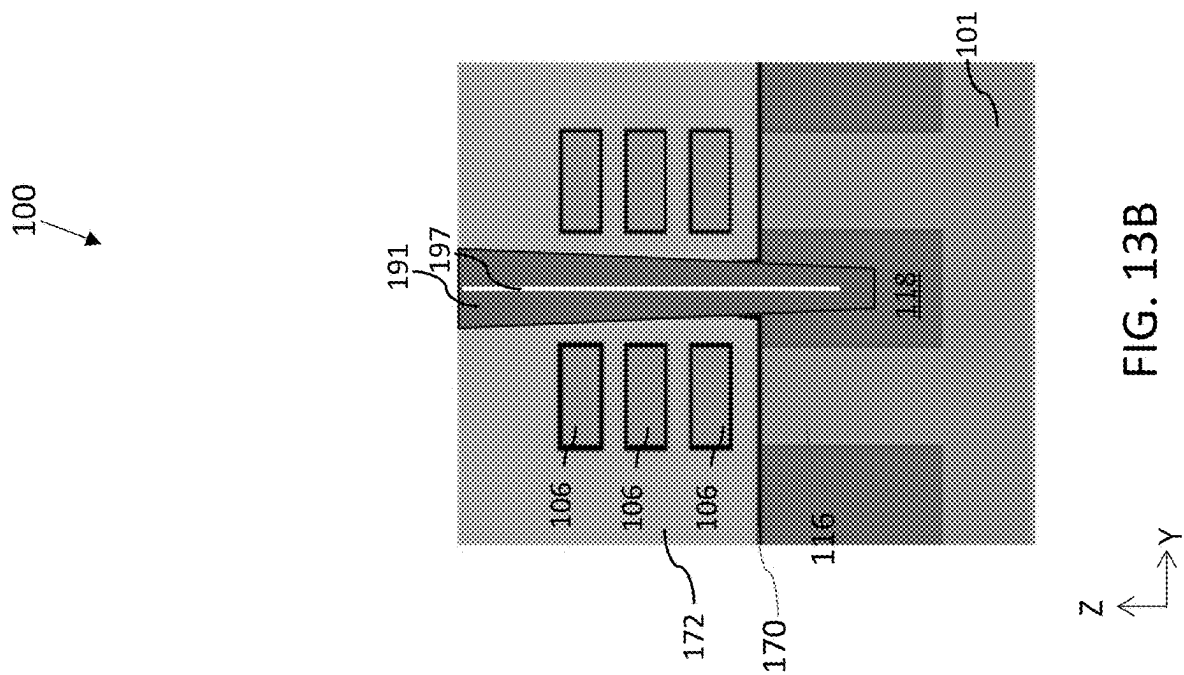
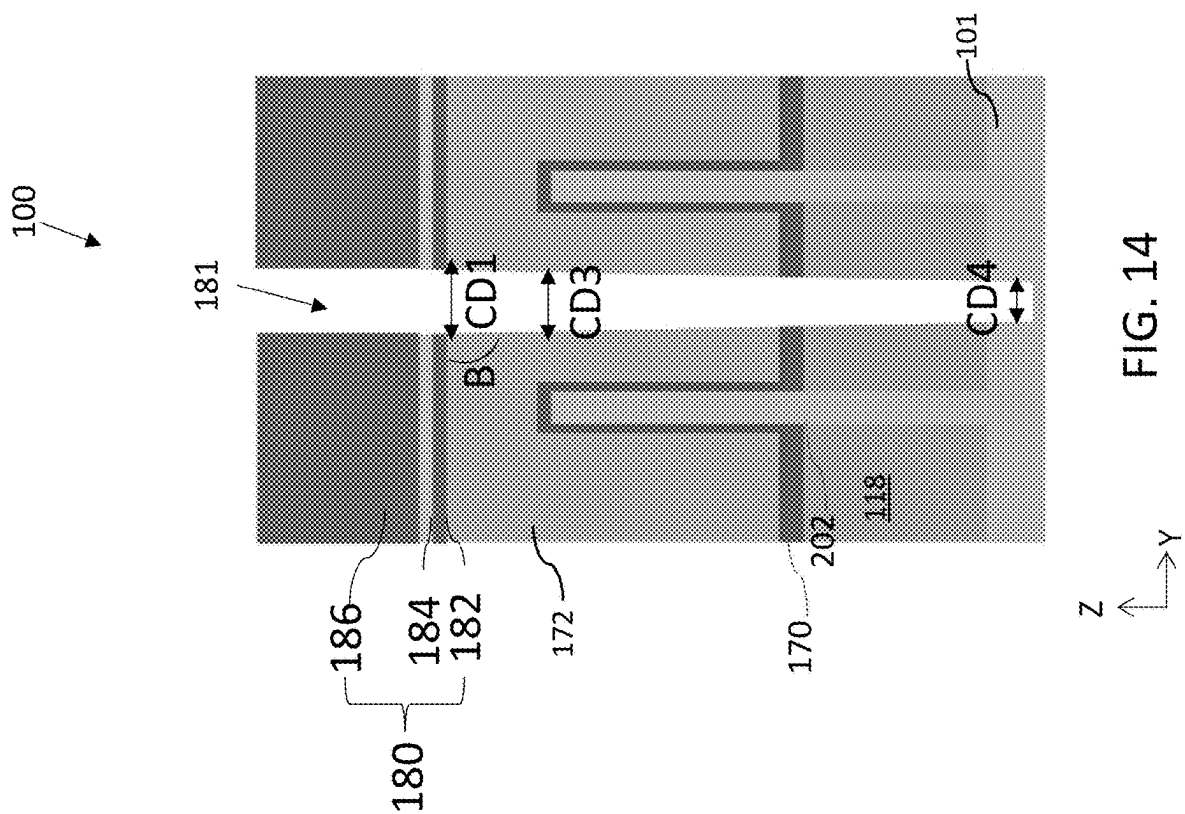


FIG. 13A





SEMICONDUCTOR DEVICE STRUCTURE AND METHODS OF FORMING THE SAME

BACKGROUND

[0001] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

[0002] Therefore, there is a need to improve processing and manufacturing ICs.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIGS. 1-6 are perspective views of various stages of manufacturing a semiconductor device structure, in accordance with some embodiments.

[0005] FIGS. 7A-11A are cross-sectional side views of various stages of manufacturing the semiconductor device structure taken along line A-A of FIG. 6, in accordance with some embodiments.

[0006] FIGS. 7B-11B are cross-sectional side views of various stages of manufacturing the semiconductor device structure taken along line B-B of FIG. 6, in accordance with some embodiments.

[0007] FIGS. 7C-11C are cross-sectional side views of various stages of manufacturing the semiconductor device structure taken along line C-C of FIG. 6, in accordance with some embodiments.

[0008] FIGS. 12A-12D are cross-sectional side views of various stages of manufacturing the semiconductor device structure taken along line B-B of FIG. 6, in accordance with some embodiments.

[0009] FIGS. 13A-13C are various views of one of various stages of manufacturing the semiconductor device structure, in accordance with some embodiments.

[0010] FIG. 14 is a cross-sectional side view of one of various stages of manufacturing the semiconductor device structure, in accordance with alternative embodiments.

DETAILED DESCRIPTION

[0011] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in

which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0012] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “on,” “top,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0013] While the embodiments of this disclosure are discussed with respect to nanostructure channel FETs, such as gate all around (GAA) FETs, for example Horizontal Gate All Around (HGAA) FETs or Vertical Gate All Around (VGAA) FETs, implementations of some aspects of the present disclosure may be used in other processes and/or in other devices, such as planar FETs, Fin-FETs, and other suitable devices. A person having ordinary skill in the art will readily understand other modifications that may be made are contemplated within the scope of this disclosure. In cases where gate all around (GAA) transistor structures are adapted, the GAA transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

[0014] FIGS. 1-13C show exemplary processes for manufacturing a semiconductor device structure **100** according to embodiments of the present disclosure. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. 1-13C, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations/processes is not limiting and may be interchangeable.

[0015] FIGS. 1-6 are perspective views of various stages of manufacturing a semiconductor device structure **100**, in accordance with some embodiments. As shown in FIG. 1, a semiconductor device structure **100** includes a stack of semiconductor layers **104** formed over a front side of a substrate **101**. The substrate **101** may be a semiconductor substrate. The substrate **101** may include a crystalline semiconductor material such as, but not limited to silicon (Si), germanium (Ge), silicon germanium (SiGe), gallium

arsenide (GaAs), indium antimonide (InSb), gallium phosphide (GaP), gallium antimonide (GaSb), indium aluminum arsenide (InAlAs), indium gallium arsenide (InGaAs), gallium antimony phosphide (GaSbP), gallium arsenic antimonide (GaAsSb) and indium phosphide (InP). In some embodiments, the substrate **101** is a silicon-on-insulator (SOI) substrate having an insulating layer (not shown) disposed between two silicon layers for enhancement. In one aspect, the insulating layer is an oxygen-containing layer.

[0016] The substrate **101** may include various regions that have been doped with impurities (e.g., dopants having p-type or n-type conductivity). Depending on circuit design, the dopants may be, for example phosphorus for an n-type field effect transistors (NFET) and boron for a p-type field effect transistors (PFET).

[0017] The stack of semiconductor layers **104** includes alternating semiconductor layers made of different materials to facilitate formation of nanostructure channels in a multi-gate device, such as nanostructure channel FETs. In some embodiments, the stack of semiconductor layers **104** includes first semiconductor layers **106** and second semiconductor layers **108**. In some embodiments, the stack of semiconductor layers **104** includes alternating first and second semiconductor layers **106**, **108**. The first semiconductor layers **106** and the second semiconductor layers **108** are made of semiconductor materials having different etch selectivity and/or oxidation rates. For example, the first semiconductor layers **106** may be made of Si and the second semiconductor layers **108** may be made of SiGe. In some examples, the first semiconductor layers **106** may be made of SiGe and the second semiconductor layers **108** may be made of Si. Alternatively, in some embodiments, either of the semiconductor layers **106**, **108** may be or include other materials such as Ge, SiC, GeAs, GaP, InP, InAs, InSb, GaAsP, AlInAs, AlGaAs, InGaAs, GaInP, GaInAsP, or any combinations thereof.

[0018] The first and second semiconductor layers **106**, **108** are formed by any suitable deposition process, such as epitaxy. By way of example, epitaxial growth of the layers of the stack of semiconductor layers **104** may be performed by a molecular beam epitaxy (MBE) process, a metalorganic chemical vapor deposition (MOCVD) process, and/or other suitable epitaxial growth processes.

[0019] The first semiconductor layers **106** or portions thereof may form nanostructure channel(s) of the semiconductor device structure **100** in later fabrication stages. The term nanostructure is used herein to designate any material portion with nanoscale, or even microscale dimensions, and having an elongate shape, regardless of the cross-sectional shape of this portion. Thus, this term designates both circular and substantially circular cross-section elongate material portions, and beam or bar-shaped material portions including, for example, a cylindrical in shape or substantially rectangular cross-section. The nanostructure channel(s) of the semiconductor device structure **100** may be surrounded by a gate electrode. The semiconductor device structure **100** may include a nanostructure transistor. The nanostructure transistors may be referred to as nanosheet transistors, nanowire transistors, gate-all-around (GAA) transistors, multi-bridge channel (MBC) transistors, or any transistors having the gate electrode surrounding the channels. The use of the first semiconductor layers **106** to define a channel or channels of the semiconductor device structure **100** is further discussed below.

[0020] Each first semiconductor layer **106** may have a thickness in a range between about 5 nm and about 30 nm. Each second semiconductor layer **108** may have a thickness that is equal, less, or greater than the thickness of the first semiconductor layer **106**. In some embodiments, each second semiconductor layer **108** has a thickness in a range between about 2 nm and about 50 nm. Three first semiconductor layers **106** and three second semiconductor layers **108** are alternately arranged as illustrated in FIG. 1, which is for illustrative purposes and not intended to be limiting beyond what is specifically recited in the claims. It can be appreciated that any number of first and second semiconductor layers **106**, **108** can be formed in the stack of semiconductor layers **104**, and the number of layers depending on the predetermined number of channels for the semiconductor device structure **100**.

[0021] In FIG. 2, fin structures **112** are formed from the stack of semiconductor layers **104**. Each fin structure **112** has an upper portion including the semiconductor layers **106**, **108** and a well portion **116** formed from the substrate **101**. The fin structures **112** may be formed by patterning a hard mask layer (not shown) formed on the stack of semiconductor layers **104** using multi-patterning operations including photo-lithography and etching processes. The etching process can include dry etching, wet etching, reactive ion etching (RIE), and/or other suitable processes. The photo-lithography process may include forming a photoresist layer (not shown) over the hard mask layer, exposing the photoresist layer to a pattern, performing post-exposure bake processes, and developing the photoresist layer to form a masking element including the photoresist layer. In some embodiments, patterning the photoresist layer to form the masking element may be performed using an electron beam (e-beam) lithography process. The etching process forms trenches **114** in unprotected regions through the hard mask layer, through the stack of semiconductor layers **104**, and into the substrate **101**, thereby leaving the plurality of extending fin structures **112**. The trenches **114** extend along the X direction. The trenches **114** may be etched using a dry etch (e.g., RIE), a wet etch, and/or combination thereof.

[0022] In FIG. 3, after the fin structures **112** are formed, an insulating material **118** is formed on the substrate **101**. The insulating material **118** fills the trenches **114** between neighboring fin structures **112** until the fin structures **112** are embedded in the insulating material **118**. Then, a planarization operation, such as a chemical mechanical polishing (CMP) method and/or an etch-back method, is performed such that the top of the fin structures **112** is exposed. The insulating material **118** may be made of silicon oxide, silicon nitride, silicon oxynitride (SiON), SiOCN, SiCN, fluorine-doped silicate glass (FSG), a low-K dielectric material, or any suitable dielectric material. The insulating material **118** may be formed by any suitable method, such as low-pressure chemical vapor deposition (LPCVD), plasma enhanced CVD (PECVD) or flowable CVD (FCVD).

[0023] In FIG. 4, the insulating material **118** is recessed to form isolation regions **120**. The recess of the insulating material **118** exposes portions of the fin structures **112**, such as the stack of semiconductor layers **104**. The recess of the insulating material **118** reveals the trenches **114** between the neighboring fin structures **112**. The isolation regions **120** may be formed using a suitable process, such as a dry etching process, a wet etching process, or a combination thereof. A top surface of the insulating material **118** may be

level with or below a surface of the second semiconductor layers **108** in contact with the well portion **116** formed from the substrate **101**.

[0024] In FIG. 5, one or more sacrificial gate structures **130** (only one is shown) are formed over the semiconductor device structure **100**. The sacrificial gate structures **130** are formed over a portion of the fin structures **112**. Each sacrificial gate structure **130** may include a sacrificial gate dielectric layer **132**, a sacrificial gate electrode layer **134**, and a mask layer **136**. The sacrificial gate dielectric layer **132**, the sacrificial gate electrode layer **134**, and the mask layer **136** may be formed by sequentially depositing blanket layers of the sacrificial gate dielectric layer **132**, the sacrificial gate electrode layer **134**, and the mask layer **136**, and then patterning those layers into the sacrificial gate structures **130**. Gate spacers **138** are then formed on sidewalls of the sacrificial gate structures **130**. The gate spacers **138** may be formed by conformally depositing one or more layers for the gate spacers **138** and anisotropically etching the one or more layers, for example. In some embodiments, the gate spacers **138** are also formed on sidewalls of the exposed portions of the fin structures **112**. While one sacrificial gate structure **130** is shown, two or more sacrificial gate structures **130** may be arranged along the X direction in some embodiments.

[0025] The sacrificial gate dielectric layer **132** may include one or more layers of dielectric material, such as a silicon oxide-based material. The sacrificial gate electrode layer **134** may include silicon such as polycrystalline silicon or amorphous silicon. The mask layer **136** may include more than one layer, such as an oxide layer and a nitride layer. The gate spacer **138** may be made of a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, SiCN, silicon oxycarbide, SiOCN, and/or combinations thereof.

[0026] The portions of the fin structures **112** that are covered by the sacrificial gate electrode layer **134** of the sacrificial gate structure **130** serve as channel regions for the semiconductor device structure **100**.

[0027] In FIG. 6, the portions of the fin structures **112** not covered by the sacrificial gate structure **130** and the gate spacers **138** are recessed to a level above, at, or below the top surfaces of the isolation regions **120**. The recess of the portions of the fin structures **112** can be done by an etch process, either isotropic or anisotropic etch process, and the etch process may be selective with respect to one or more crystalline planes of the substrate **101**. The etch process may be a dry etch, such as a RIE, NBE, or the like, or a wet etch, such as using tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH₄OH), or any suitable etchant.

[0028] FIGS. 7A, 7B, and 7C are cross-sectional side views of the semiconductor device structure **100** taken along line A-A, line B-B, and line C-C of FIG. 6, respectively.

[0029] FIGS. 8A, 8B, and 8C are cross-sectional side views of one of various stages of manufacturing the semiconductor device structure **100** taken along line A-A, line B-B, and line C-C of FIG. 6, respectively, in accordance with some embodiments. As shown in FIG. 8A, edge portions of each second semiconductor layer **108** of the stack of semiconductor layers **104** are removed horizontally along the X direction. The removal of the edge portions of the second semiconductor layers **108** forms cavities. In some embodiments, the portions of the second semiconductor layers **108** are removed by a selective wet etch process. In

cases where the second semiconductor layers **108** are made of SiGe and the first semiconductor layers **106** are made of silicon, the second semiconductor layer **108** can be selectively etched using a wet etchant such as, but not limited to, ammonium hydroxide (NH₄OH), tetramethylammonium hydroxide (TMAH), ethylenediamine pyrocatechol (EDP), or potassium hydroxide (KOH) solutions.

[0030] After removing edge portions of each second semiconductor layers **108**, a dielectric layer is deposited in the cavities to form dielectric spacers **144**. The dielectric spacers **144** may be made of a low-K dielectric material, such as SiON, SiCN, SiOC, SiOCN, or SiN. The dielectric spacers **144** may be formed by first forming a conformal dielectric layer using a conformal deposition process, such as ALD, followed by an anisotropic etching to remove portions of the conformal dielectric layer other than the dielectric spacers **144**. The dielectric spacers **144** are protected by the first semiconductor layers **106** during the anisotropic etching process. The remaining second semiconductor layers **108** are capped between the dielectric spacers **144** along the X direction.

[0031] FIGS. 9A, 9B, and 9C are cross-sectional side views of one of various stages of manufacturing the semiconductor device structure **100** taken along line A-A, line B-B, and line C-C of FIG. 6, respectively, in accordance with some embodiments. As shown in FIGS. 9A and 9C, source/drain (S/D) regions **146** are formed from the well portion **116**. The S/D regions **146** may grow both vertically and horizontally to form facets, which may correspond to crystalline planes of the material used for the well portion **116**. In this disclosure, a source region and a drain region are interchangeably used, and the structures thereof are substantially the same. Furthermore, source/drain region(s) may refer to a source or a drain, individually or collectively dependent upon the context. The S/D regions **146** may be made of one or more layers of Si, SiP, SiC and SiCP for n-channel FETs or Si, SiGe, Ge for p-channel FETs. For p-channel FETs, p-type dopants, such as boron (B), may also be included in the S/D regions **146**. The S/D regions **146** may be formed by an epitaxial growth method using CVD, ALD or MBE.

[0032] FIGS. 10A, 10B, and 10C are cross-sectional side views of one of various stages of manufacturing the semiconductor device structure **100** taken along line A-A, line B-B, and line C-C of FIG. 6, respectively, in accordance with some embodiments. In FIGS. 10A, 10B, and 10C, a contact etch stop layer (CESL) **162** is conformally formed on the exposed surfaces of the semiconductor device structure **100**. The CESL **162** covers the sidewalls of the sacrificial gate structure **130**, the insulating material **118**, and the S/D regions **146**. The CESL **162** may include an oxygen-containing material or a nitrogen-containing material, such as silicon nitride, silicon carbon nitride, silicon oxynitride, carbon nitride, silicon oxide, silicon carbon oxide, or the like, or a combination thereof, and may be formed by CVD, PECVD, ALD, or any suitable deposition technique. Next, an interlayer dielectric (ILD) layer **164** is formed on the CESL **162** over the semiconductor device structure **100**. The materials for the ILD layer **164** may include compounds including Si, O, C, and/or H, such as silicon oxide, SiCOH, or SiOC. Organic materials, such as polymers, may also be used for the ILD layer **164**. The ILD layer **164** may be deposited by a PECVD process or other suitable deposition technique. In some embodiments, after formation of the ILD

layer **164**, the semiconductor device structure **100** may be subject to a thermal process to anneal the ILD layer **164**.

[0033] After the ILD layer **164** is formed, a planarization operation, such as CMP, is performed on the semiconductor device structure **100** until the sacrificial gate electrode layer **134** is exposed, as shown in FIGS. **10A** and **10B**.

[0034] FIGS. **11A**, **11B**, and **11C** are cross-sectional side views of one of various stages of manufacturing the semiconductor device structure **100** taken along line A-A, line B-B, and line C-C of FIG. **6**, respectively, in accordance with some embodiments. As shown in FIGS. **11A** and **11B**, the sacrificial gate structure **130** and the second semiconductor layers **108** are removed. The removal of the sacrificial gate structure **130** and the semiconductor layers **108** forms an opening between gate spacers **138** and between first semiconductor layers **106**. The ILD layer **164** protects the S/D regions **146** during the removal processes. The sacrificial gate structure **130** can be removed using plasma dry etching and/or wet etching. The sacrificial gate electrode layer **134** may be first removed by any suitable process, such as dry etch, wet etch, or a combination thereof, followed by the removal of the sacrificial gate dielectric layer **132**, which may also be performed by any suitable process, such as dry etch, wet etch, or a combination thereof. In some embodiments, a wet etchant such as a tetramethylammonium hydroxide (TMAH) solution can be used to selectively remove the sacrificial gate electrode layer **134** but not the gate spacers **138**, the ILD layer **164**, and the CESL **162**.

[0035] Portions of the second semiconductor layers **108** may be removed using a selective wet etching process. In cases where the second semiconductor layers **108** are made of SiGe and the first semiconductor layers **106** are made of Si, the chemistry used in the selective wet etching process removes the SiGe while not substantially affecting Si, the dielectric materials of the gate spacers **138**, and the dielectric spacers **144**. In one embodiment, the second semiconductor layers **108** can be removed using a wet etchant such as, but not limited to, hydrofluoric (HF), nitric acid (HNO₃), hydrochloric acid (HCl), phosphoric acid (H₃PO₄), a dry etchant such as fluorine-based (e.g., F₂) or chlorine-based gas (e.g., Cl₂), or any suitable isotropic etchants.

[0036] After the formation of the nanostructure channels (i.e., the exposed portions of the first semiconductor layers **106**), a gate dielectric layer **170** is formed to surround the exposed portions of the first semiconductor layers **106**, and a gate electrode layer **172** is formed on the gate dielectric layer **170**. The gate dielectric layer **170** and the gate electrode layer **172** may be collectively referred to as a gate structure **174**. In some embodiments, an interfacial layer (IL) (not shown) is formed between the gate dielectric layer **170** and the exposed surfaces of the first semiconductor layers **106**. In some embodiments, the gate dielectric layer **170** includes one or more layers of a dielectric material, such as silicon oxide, silicon nitride, or high-K dielectric material, other suitable dielectric material, and/or combinations thereof. Examples of high-K dielectric material include HfO₂, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina (HfO₂—Al₂O₃) alloy, other suitable high-K dielectric materials, and/or combinations thereof. The gate dielectric layer **170** may be formed by CVD, ALD or any suitable deposition technique. The gate electrode layer **172** may include one or more layers of conductive material, such as polysilicon, aluminum, copper, titanium, tantalum, tungsten,

cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, metal alloys, other suitable materials, and/or any combinations thereof. The gate electrode layer **172** may be formed by CVD, ALD, electro-plating, or other suitable deposition technique. The gate electrode layer **172** may be also deposited over the upper surface of the ILD layer **164**. The gate dielectric layer **170** and the gate electrode layer **172** formed over the ILD layer **164** are then removed by using, for example, CMP, until the top surface of the ILD layer **164** is exposed.

[0037] FIGS. **12A-12D** are cross-sectional side views of various stages of manufacturing the semiconductor device structure **100** taken along line B-B of FIG. **6**, in accordance with some embodiments. As shown in FIG. **12A**, a mask structure **180** is formed on the gate electrode layer **172** and the ILD layer **164**, and a resist structure **190** is formed on the mask structure **180**. In some embodiments, the mask structure **180** includes a first dielectric layer **182**, a semiconductor layer **184** deposited on the first dielectric layer, and a second dielectric layer **186** deposited on the semiconductor layer **184**. The first and second dielectric layers **182**, **186** may include any suitable dielectric material. In some embodiments, the first and second dielectric layers **182**, **186** are SiN layers. The semiconductor layer **184** may be any suitable semiconductor layer. In some embodiments, the semiconductor layer **184** is an amorphous silicon layer. In some embodiments, the resist structure **190** is a tri-layer photoresist. For example, the resist structure **190** may include a bottom layer **192** deposited on the mask structure **180**, a middle layer **194** deposited on the bottom layer **192**, and a photoresist layer **196** deposited on the middle layer **194**. The bottom layer **192** and the middle layer **194** are made of different materials such that the optical properties and/or etching properties of the bottom layer **192** and the middle layer **194** are different from each other. In some embodiments, the bottom layer **192** may be a carbon layer, and the middle layer **194** may be a silicon-rich layer designed to provide an etch selectivity between the middle layer **194** and the bottom layer **192**. The photoresist layer **196** may be a chemically amplified photoresist layer and can be a positive tone photoresist or a negative tone photoresist. The photoresist layer **196** may include a polymer, such as phenol formaldehyde resin, a poly(norbornene)-co-maleic anhydride (COMA) polymer, a poly(4-hydroxystyrene) (PHS) polymer, a phenol-formaldehyde (bakelite) polymer, a polyethylene (PE) polymer, a polypropylene (PP) polymer, a polycarbonate polymer, a polyester polymer, or an acrylate-based polymer, such as a poly (methyl methacrylate) (PMMA) polymer or poly (methacrylic acid) (PMAA). The photoresist layer **196** may be formed by spin-on coating. The photoresist layer **196** may be patterned to have an opening **181** formed therein.

[0038] As shown in FIG. **12B**, the opening **181** is extended into the middle layer **194**, the bottom layer **192**, and the mask structure **180**. The opening **181** may be extended into the middle layer **194**, the bottom layer **192**, and the mask structure **180** by one or more etch processes. In some embodiments, the resist structure **190** is removed during the one or more etch processes to extend the opening **181** into the mask structure **180**. In some embodiments, the resist structure **190** is removed after the opening **181** is extended into the mask structure **180**. A portion of the gate electrode layer **172** is exposed in opening **181**.

[0039] As shown in FIG. 12C, the exposed portion of the gate electrode layer 172 is removed. The exposed portion of the gate electrode layer 172 may be removed by any suitable method. In some embodiments, a cyclic process is performed to remove the exposed portion of the gate electrode layer 172. The cyclic process includes a plurality of cycles, and each cycle includes depositing a layer on the exposed portion of the gate electrode layer 172, oxidizing the layer to form an oxide layer, removing portions of the oxide layer, and removing a portion of the exposed portion of the gate electrode layer 172. In some embodiments, the layer may be a silicon-containing layer and may be formed by any suitable process, such as CVD or PECVD. A silicon-containing precursor, such as SiCl_4 , may be introduced into the processing chamber in which the semiconductor device structure 100 is disposed therein, and the silicon-containing layer is formed on the exposed surfaces of the semiconductor device structure 100. In some embodiments, the silicon-containing layer is formed on the second dielectric layer 186, the sidewalls of the first dielectric layer 182, the semiconductor layer 184, and the second dielectric layer 186 in the opening 181, and the exposed surface of the gate electrode layer 172. Next, the oxidation process is performed to oxidize the silicon-containing layer to form the oxide layer, such as a silicon oxide layer. The oxidation process may be any suitable oxidation process, such as a thermal oxidation process. Next, a breakthrough etch process is performed to remove portions of the oxide layer formed on horizontal surfaces of the semiconductor device structure 100, such as the portions of the oxide layer formed on the second dielectric layer 186 and the exposed surface of the gate electrode layer 172. The breakthrough etch process may be an anisotropic dry etch process, such as a plasma etch process. In some embodiments, a fluorine-containing etchant, such as C_4F_6 , may be used in the anisotropic dry etch process. After the breakthrough etch process, the portion of the gate electrode layer 172 located at the bottom of the opening 181 is exposed, and a main etch process is performed to remove a portion of the gate electrode layer 172. In some embodiments, the main etch process is a dry etch process, such as an anisotropic plasma etch process. One of more etchants may be used in the main etch process. In some embodiments, one or more chlorine-containing etchants, such as BCl_3 and/or Cl_2 , may be used. The oxide layer formed on the sidewalls in the opening 181 may be also removed by the one or more chlorine-containing etchants. In some embodiments, a deposition gas, such as a silicon-containing gas, for example SiCl_4 , may be introduced into the processing chamber along with the etchants. The deposition gas may form a layer on sidewalls of the gate electrode layer 172 in the opening 181 as the opening 181 extends into the gate electrode layer 172. The layer protects the sidewalls of the gate electrode layer 172 during the main etch process to protect the sidewalls of the gate electrode layer 172. In other words, the oxide layer formed on the sidewalls in the opening 181 may be removed by the etchants of the main etch process, while the deposition gas of the main etch process forms the layer, which may be a silicon layer, on the sidewalls in the opening 181. In some embodiments, the time duration of the main etch process may be substantially short, such as from about 10 seconds to about 20 seconds.

[0040] The cycle of depositing a layer on the exposed portion of the gate electrode layer 172, oxidizing the layer

to form an oxide layer, removing portions of the oxide layer, and removing a portion of the exposed portion of the gate electrode layer 172 is repeated multiple times so the opening 181 extends through the gate electrode layer 172 and into the insulating material 118, as shown in FIG. 12C. After the opening 181 is first extended into the gate electrode layer 172, the depositing a layer on the exposed portion of the gate electrode layer 172 and oxidizing the layer to form an oxide layer of the cyclic process form the oxide layer on the sidewalls of the gate electrode layer 172 in the opening 181 to protect the sidewalls of the gate electrode layer 172. In some embodiments, the number of cycles ranges from about eight to about 15, such as 12. At the end of the cycles, the opening 181 extends through the gate electrode layer 172. In some embodiments, the opening 181 extends through the gate dielectric layer 170 and the insulating material 118 and into the substrate 101, as shown in FIG. 12C. In some embodiments, the opening 181 is about 5 nm to about 35 nm into the substrate 101.

[0041] In some embodiments, one or more wet clean processes may be performed in between the plurality of cycles to remove byproducts formed from the etching processes. For example, in some embodiments, after seven cycles, a wet clean process is performed. The wet clean process does not substantially affect the materials of the semiconductor device structure 100, such as the gate electrode layer 172 and the gate dielectric layer 170. The wet clean process is performed after a first number of cycles are performed, and a second number of cycles are performed after the wet clean process. In some embodiments, the first number of cycles is greater than the second number of cycles. For example, the first number of cycles includes seven cycles, and the second number of cycles includes five cycles. In some embodiments, each cycle of the first number of cycles includes the main etch process with a first plasma bias voltage, and each cycle of the second number of cycles includes the main etch process with a second plasma bias voltage substantially greater than the first plasma bias voltage. For example, the main etch process of the first number of cycles has a first plasma bias voltage ranging from about 150 V to about 250 V, and the main etch process of the second number of cycles has a second plasma bias voltage ranging from about 800 V to about 1200 V. In some embodiments, the second plasma bias voltage is about three times to about seven times the first plasma bias voltage. The higher plasma bias voltage may lead to a more anisotropic etch process that decreases CD3 but increases CD1 and CD2. With higher bias voltage, CD2 may be increased due to more plasma bombardment. In some embodiments, the critical dimension CD1 is substantially greater than a critical dimension CD2 of the opening 181 at the top of the mask structure 180 and substantially less than a critical dimension CD3 of the opening 181 at a location below the portion of the opening 181 having the critical dimension CD1, as shown in FIG. 12C. In some embodiments, the critical dimension CD3 is at a location adjacent the top surface of the topmost first semiconductor layer 106, as shown in FIG. 12C. Because the main etch process is substantially selective (i.e., the gate electrode layer 172 is etched at a substantially faster rate than the materials of the mask structure 180), the critical dimension CD1 is substantially smaller than the critical dimension CD2. As described above, the second plasma bias voltage is substantially greater than the first

plasma bias voltage. As a result, the critical dimension CD3 is substantially greater than the critical dimension CD1.

[0042] As shown in FIG. 12D, a trim process may be performed to remove portions of the mask structure 180, so the opening 181 in the mask structure 180 is widened. In other words, the trim process enlarges the critical dimension CD2. In some embodiments, a plasma etch process is performed to trim the mask structure 180. The plasma etch process may utilize a fluorine-containing etchant, such as C_4F_6 . In some embodiments, the critical dimension CD1 is also enlarged by the trim process. If the critical dimension CD1 gets too large, the conductive contact to be formed on the gate electrode layer 172 may land on a portion of the dielectric material 191 (FIG. 13B) formed in the opening 181, leading to increased contact resistance. In some embodiments, in order to prevent the critical dimension CD1 from getting too large, a carbon-containing gas, such as CH_4 , is introduced along with the fluorine-containing etchant during the plasma etch process. The carbon-containing gas may form a polymer layer on the sidewalls of the opening 181. As a result, an angle A formed between the sidewall of the mask structure 180 exposed to the opening 181 and the top surface of the gate electrode layer 172 is greater than about 75 degrees, such as from about 75 degrees to about 90 degrees. If the angle A is greater than about 90 degrees, the critical dimension CD2 may be too small, and a large seam or void may form in the dielectric material 191 (FIG. 13B). On the other hand, if the angle A is less than about 75 degrees, the top portion of the gate electrode layer 172 exposed to the opening 181 may be removed, leading to enlarged critical dimension CD1. In some embodiments, an angle B is formed between the sidewall of the gate electrode layer 172 exposed in the opening 181 and the top surface of the gate electrode layer 172, and the angle B ranges from about 90 degrees to about 95 degrees. In some embodiments, the angle B is substantially greater than the angle A.

[0043] In some embodiments, the polymer layer is formed on the sidewalls of the mask structure 180, the gate electrode layer 172, and the insulating material 118 exposed to the opening 181. A plasma process may be performed to remove the polymer layer. In some embodiments, the plasma process is an oxygen plasma burn process, in which the polymer layer is removed by the oxygen-containing plasma. The plasma process is different from the trim process.

[0044] After the trim process and the plasma process, a wet clean process may be performed to remove any byproducts from the trim process. In some embodiments, after the wet clean process, the critical dimension CD 1 may range from about 19 nm to about 26 nm, and the critical dimension CD3 may range from about 19 nm to about 26 nm. In some embodiments, the critical dimension CD3 is less than the critical dimension CD1. A critical dimension CD4 located at the bottom of the opening 181 may range from about 12 nm to about 16 nm. In some embodiments, a width of the first semiconductor layer 106 along the Y-direction ranges from about 20 nm to about 40 nm. In some embodiments, the width of the first semiconductor layer 106 may be greater than the critical dimension CD1. In some embodiments, the width of the first semiconductor layer 106 may be smaller than the critical dimension CD1.

[0045] FIGS. 13A-13C are various views of one of various stages of manufacturing the semiconductor device structure 100, in accordance with some embodiments. As shown in FIGS. 13A-13C, a dielectric material 191 is formed in the

opening 181, the mask structure 180 is removed, and conductive features 193 are formed in the ILD layer 164. Silicide layers 195 may be formed between S/D regions 146 and the conductive features 193. The dielectric material 191 may include any suitable dielectric material, such as SiN, HfO, $SiNO_x$, or ZrO. The conductive feature 193 may include any suitable electrically conductive material, such as a metal. The silicide layer 195 may be a metal silicide layer that is a result of a reaction between the metal of the conductive feature 193 and the semiconductor material of the S/D region 146. In some embodiments, a seam 197 is formed in the dielectric material 191. Because the critical dimension CD1 is greater than the critical dimension CD3, the seam 197 may have a dimension along the Y-direction less than about 1 nm.

[0046] FIG. 14 is a cross-sectional side view of one of various stages of manufacturing the semiconductor device structure 100, in accordance with alternative embodiments. As described above, the semiconductor device structure 100 may include nanostructure channel FETs. In some embodiments, the semiconductor device structure 100 includes FinFETs. As shown in FIG. 14, the semiconductor device structure 100 includes one or more fin structures 202 extending from the substrate 101, the gate dielectric layer 170 formed around the fin structures 202, and the gate electrode layer 172 formed on the gate dielectric layer 170. The opening 181 is formed in the mask structure 180, the gate electrode layer 172, the insulating material 118, and into the substrate 101. In some embodiments, a width of the fin structure 202 along the Y-direction ranges from about 5 nm to about 8 nm, and the critical dimension CD4 is substantially greater than the width of the fin structure 202.

[0047] Embodiments of the present disclosure provide a method to form a semiconductor device structure 100. The method includes forming an opening through a gate electrode layer 172 by a cyclic process and performing a trim process to remove portions of a mask structure 180. The cyclic process includes a first plurality of cycles, a second plurality of cycles, and a clean process separating the first and second pluralities of cycles. A main etch process of the first plurality of cycles includes a first plasma bias voltage substantially smaller than a second plasma bias voltage of a main etch process of the second plurality of cycles. Some embodiments may achieve advantages. For example, with the increased plasma bias voltage, the critical dimension CD1 of the portion of the opening 181 located at the top of the gate electrode layer 172 remains to be small. Furthermore, the trim process includes a carbon-containing gas to form a polymer layer on sidewalls of the opening 181, and the critical dimension CD1 is not unnecessarily enlarged.

[0048] An embodiment is a method. The method includes forming a fin structure over a substrate, forming an insulating material adjacent the fin structure, depositing a gate dielectric layer over the fin structure and the insulating material, depositing a gate electrode layer on the gate dielectric layer, and forming an opening through the gate electrode layer and the gate dielectric layer into the insulating material by a cyclic process. The cyclic process includes performing a first number of cycles, each cycle of the first number of cycles includes a first main etch process having a first plasma bias voltage, performing a second number of cycles, and each cycle of the second number of cycles includes a second main etch process having a second plasma

bias voltage substantially greater than the first plasma bias voltage. The method further includes filling the opening with a dielectric material.

[0049] Another embodiment is a method. The method includes forming an insulating material over a substrate, depositing a gate dielectric layer on the insulating material, depositing a gate electrode layer on the gate dielectric layer, forming a mask structure on the gate electrode layer, performing a cyclic process to form an opening through the mask structure and the gate electrode layer, and performing a trim process to remove portions of the mask structure. The trim process includes introducing an etchant and a carbon-containing gas. The method further includes filling the opening with a dielectric material.

[0050] A further embodiment is a method. The method includes forming an insulating material over a substrate, depositing a gate dielectric layer on the insulating material, depositing a gate electrode layer on the gate dielectric layer, forming a mask structure on the gate electrode layer, and performing a cyclic process to form an opening through the mask structure and the gate electrode layer. The opening has a first critical dimension at a top of the mask structure, a second critical dimension at a top of the gate electrode layer, a third critical dimension at a location below the second critical dimension, and the second critical dimension is substantially greater than the first critical dimension and substantially less than the third critical dimension. The method further includes performing a trim process to remove portions of the mask structure. The first critical dimension is substantially greater than the second critical dimension, and the second critical dimension is substantially greater than the third critical dimension after the trim process. The method further includes filling the opening with a dielectric material.

[0051] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

1. A method, comprising:

forming a fin structure over a substrate;
forming an insulating material adjacent the fin structure;
depositing a gate dielectric layer over the fin structure and the insulating material;
depositing a gate electrode layer on the gate dielectric layer;
forming an opening through the gate electrode layer and the gate dielectric layer into the insulating material by a cyclic process, wherein the cyclic process comprises:
performing a first number of cycles, wherein each cycle of the first number of cycles includes a first main etch process having a first plasma bias voltage; and
performing a second number of cycles, wherein each cycle of the second number of cycles includes a

second main etch process having a second plasma bias voltage substantially greater than the first plasma bias voltage; and

filling the opening with a dielectric material.

2. The method of claim 1, wherein the fin structure comprises a plurality of semiconductor layers, and the gate dielectric layer and the gate electrode layer surround a portion of each semiconductor layer of the plurality of semiconductor layers.

3. The method of claim 1, wherein the first plasma bias voltage ranging from about 150 V to about 250 V, and the second plasma bias voltage ranging from about 800 V to about 1200 V.

4. The method of claim 1, wherein the first number of cycles includes more cycles than the second number of cycles.

5. The method of claim 1, wherein each cycle of the first number of cycles comprises:

depositing a layer on the gate electrode layer;
oxidizing the layer to form an oxide layer;
removing a portion of the oxide layer; and
performing the first main etch process to remove a portion of the gate electrode layer.

6. The method of claim 5, wherein the first main etch process uses one or more chlorine-containing etchants.

7. The method of claim 6, wherein the one or more chlorine-containing etchants comprise BCl_3 and Cl_2 .

8. The method of claim 1, further comprising performing a clean process between the first number of cycles and the second number of cycles.

9. A method, comprising:

forming an insulating material over a substrate;
depositing a gate dielectric layer on the insulating material;
depositing a gate electrode layer on the gate dielectric layer;
forming a mask structure on the gate electrode layer;
performing a cyclic process to form an opening through the mask structure and the gate electrode layer;
performing a trim process to remove portions of the mask structure, wherein the trim process comprises introducing an etchant and a carbon-containing gas; and
filling the opening with a dielectric material.

10. The method of claim 9, wherein the trim process is a plasma etch process.

11. The method of claim 10, wherein a polymer layer is formed on sidewalls of the opening during the trim process.

12. The method of claim 11, further comprising a plasma process to remove the polymer layer after the trim process.

13. The method of claim 12, wherein the plasma process forms an oxygen-containing plasma.

14. A method, comprising:

forming an insulating material over a substrate;
depositing a gate dielectric layer on the insulating material;
depositing a gate electrode layer on the gate dielectric layer;
forming a mask structure on the gate electrode layer;
performing a cyclic process to form an opening through the mask structure and the gate electrode layer, wherein the opening has a first critical dimension at a top of the mask structure, a second critical dimension at a top of the gate electrode layer, a third critical dimension at a location below the second critical dimension, and the

second critical dimension is substantially greater than the first critical dimension and substantially less than the third critical dimension;

performing a trim process to remove portions of the mask structure, wherein the first critical dimension is substantially greater than the second critical dimension, and the second critical dimension is substantially greater than the third critical dimension after the trim process; and filling the opening with a dielectric material.

15. The method of claim **14**, wherein a first angle is formed between a sidewall of the mask structure exposed in the opening and a top surface of the gate electrode layer after the trim process, and the first angle ranges from about 75 degrees to about 90 degrees.

16. The method of claim **15**, wherein a second angle is formed between a sidewall of the gate electrode layer

exposed in the opening and the top surface of the gate electrode layer after the trim process, and the second angle ranges from about 90 degrees to about 95 degrees.

17. The method of claim **16**, wherein the second angle is substantially greater than the first angle.

18. The method of claim **14**, further comprising forming a plurality of semiconductor layers, wherein the gate dielectric layer and the gate electrode layer surrounds the plurality of semiconductor layers.

19. The method of claim **14**, further comprising forming a fin structure, wherein the gate dielectric layer and the gate electrode layer are disposed around the fin structure.

20. The method of claim **14**, further comprising performing a plasma process to remove a polymer layer from sidewalls of the openings after the trim process.

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